

Hypothesis Testing II



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Example: Two-Tail Test



The average cost of a blood test in a hospital is said to be Tk.168. A random sample of 25 patients resulted in $\bar{x} = \text{Tk.}172.50$ and $s = \text{Tk.}15.40$. Test at the $\alpha = 0.05$ level.

(Assume the population distribution is normal)



$$H_0: \mu = 168$$

$$H_1: \mu \neq 168$$



Example Solution: Two-Tail Test

$$H_0: \mu = 168$$

$$H_1: \mu \neq 168$$

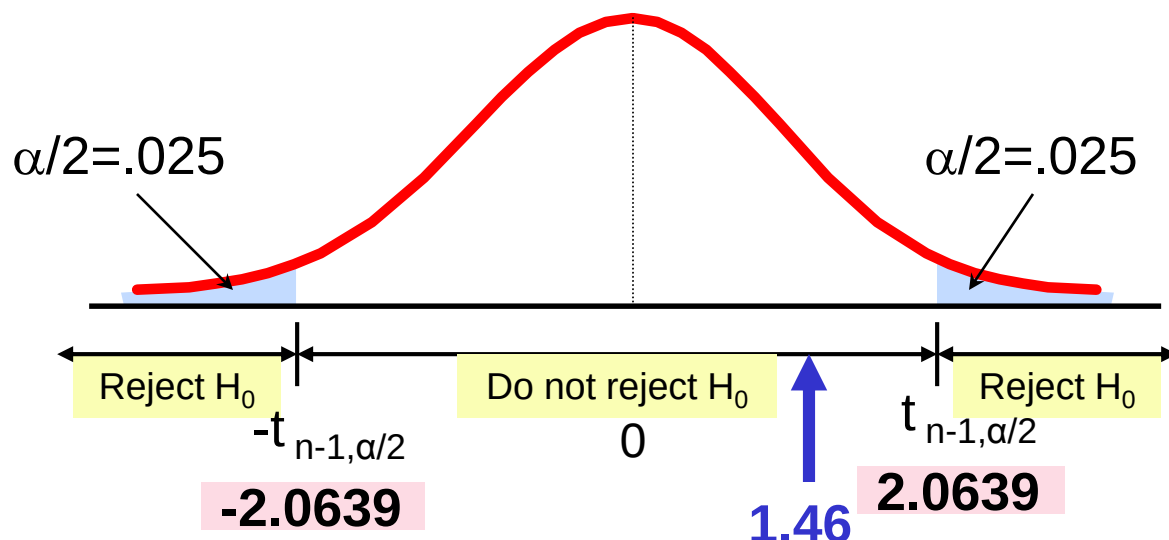
■ $\alpha = 0.05$

■ $n = 25$

■ σ is unknown, so
use a **t statistic**

■ Critical Value:

$$t_{24, .025} = \pm 2.0639$$



$$t_{n-1} = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{172.50 - 168}{\frac{15.40}{\sqrt{25}}} = 1.46$$

Do not reject H_0 : not sufficient evidence that true mean cost is different than Tk.168



Matched Pairs

Matched Pairs

Tests Means of 2 **Related** Populations

- Paired or matched samples
- Repeated measures (before/after)
- Use **difference** between paired values:

$$d_i = x_i - y_i$$

- Assumptions:
 - Both Populations Are Normally Distributed



Test Statistic: Matched Pairs

Matched
Pairs

The test statistic for the mean difference is a **t value**, with **$n - 1$ degrees of freedom**:

$$t = \frac{\bar{d} - D_0}{\frac{s_d}{\sqrt{n}}}$$

Where

D_0 = hypothesized mean difference

s_d = sample standard dev. of differences

n = the sample size (number of pairs)



Matched Pairs Example

- Assume you send your Teachers to a training workshop. Has the training made a difference in the number of complaints? You collect the following data:

<u>Teachers</u>	<u>Number of Complaints:</u>		<u>(2) - (1)</u> <u>Difference, d_i</u>
	<u>Before (1)</u>	<u>After (2)</u>	
Uni1	6	4	- 2
Uni2	20	6	-14
Uni3	3	2	- 1
Uni4	0	0	0
Uni5	4	0	- 4
			<u>-21</u>

$$\begin{aligned}\bar{d} &= \frac{\sum d_i}{n} \\ &= - 4.2\end{aligned}$$

$$\begin{aligned}S_d &= \sqrt{\frac{\sum (d_i - \bar{d})^2}{n-1}} \\ &= 5.67\end{aligned}$$



Matched Pairs: Solution

■ Has the training made a difference in the number of complaints (at the $\alpha = 0.01$ level)?

$$H_0: \mu_x - \mu_y = 0$$

$$H_1: \mu_x - \mu_y \neq 0$$

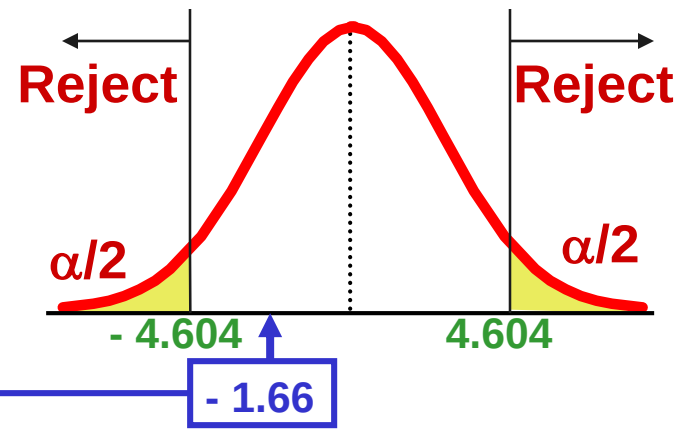
$$\alpha = .01 \quad \bar{d} = -4.2$$

Critical Value = ± 4.604

$$\text{d.f.} = n - 1 = 4$$

Test Statistic:

$$t = \frac{\bar{d} - D_0}{s_d / \sqrt{n}} = \frac{-4.2 - 0}{5.67 / \sqrt{5}} = -1.66$$



Decision: Do not reject H_0
(t stat is not in the reject region)

Conclusion: There is not a significant change in the number of complaints.



Difference Between Two Means

Population means,
independent
samples

Goal: Form a confidence interval
for the difference between two
population means, $\mu_x - \mu_y$

- Different data sources
 - Unrelated
 - Independent
 - Sample selected from one population has no effect on the sample selected from the other population



Pooled Variance t Test: Example

You are a financial analyst for a brokerage firm. Is there a difference in dividend yield between stocks listed on the DSE & CSE? You collect the following data:

	<u>DSE</u>	<u>CSE</u>
Number	21	25
Sample mean	3.27	2.53
Sample std dev	1.30	1.16

Assuming both populations are approximately normal with equal variances, is there a difference in average yield ($\alpha = 0.05$)?





Calculating the Test Statistic

The test statistic is:

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{(3.27 - 2.53) - 0}{\sqrt{1.5021 \left(\frac{1}{21} + \frac{1}{25} \right)}} = 2.040$$

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{(n_1 - 1) + (n_2 - 1)} = \frac{(21 - 1)1.30^2 + (25 - 1)1.16^2}{(21 - 1) + (25 - 1)} = 1.5021$$



Solution

$$H_0: \mu_1 - \mu_2 = 0 \text{ i.e. } (\mu_1 = \mu_2)$$

$$H_1: \mu_1 - \mu_2 \neq 0 \text{ i.e. } (\mu_1 \neq \mu_2)$$

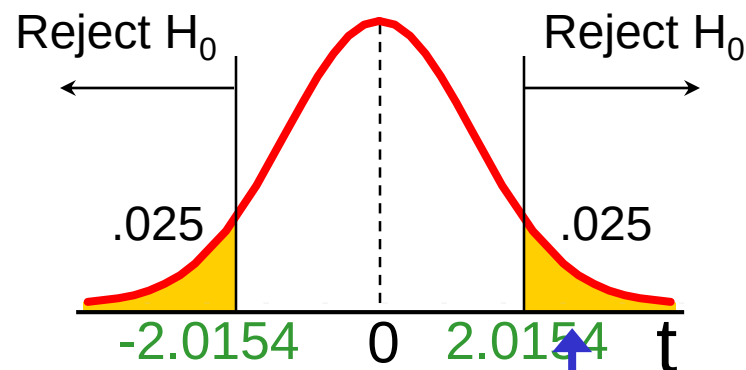
$$\alpha = 0.05$$

$$df = 21 + 25 - 2 = 44$$

$$\text{Critical Values: } t = \pm 2.0154$$

Test Statistic:

$$t = \frac{3.27 - 2.53}{\sqrt{1.5021 \left(\frac{1}{21} + \frac{1}{25} \right)}} = 2.040$$



2.040

Decision:

Reject H_0 at $\alpha = 0.05$

Conclusion:

There is evidence of a difference in means.